Highlights

- Oral peptide enhancers act in a micro-niche; a mechanistic model quantifies it
- Paired Monte Carlo arms isolate what co-localization adds to bioavailability
- Matrix-only gain survives uptake limits; it is thinning plus protease shield
- Optimized release is tested out of sample, not reported as in-sample optimum
- Failure boundaries give measurable pass-fail criteria for the first experiments